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Is it alpha or beta? Decomposing hedge fund returns when models are misspecified

Contents

1 Big picture

- **Question.** How should hedge fund returns be decomposed into alpha and beta when all practical factor models are misspecified?
- **Core problem.** If a model omits alternative strategies followed by hedge funds, the omitted beta premia show up as alpha. The apparent skill of a fund may therefore be hidden systematic exposure.
- **Method.** The paper builds a nonparametric approach that estimates the entire cross-sectional distributions of alpha and beta components for any chosen model and provides formal tests for comparing misspecified models.
- **Data.** The empirical analysis uses monthly net-of-fee returns for 5,231 hedge funds from January 1994 to December 2020, combining BarclayHedge, HFR, Morningstar, and TASS.
- **Models compared.** The paper studies nine models: CAPM, four standard models, two machine-learning models, and two economically motivated hedge-fund models with additional factors.
- **Main finding.** Standard models and equity-trained machine-learning models are as misspecified as the CAPM for hedge funds. They leave alternative-strategy premia inside alpha.
- **Best-performing model.** The CP model, using market, size, illiquidity, BAB, variance, carry, and time-series momentum, sharply lowers measured alpha and raises measured beta.
- **Key numbers.** Under the CAPM, the average alpha is 2.93% per year and 72.85% of funds have positive alpha. Under CP, the average alpha falls to 0.37% and 53.03% of funds have positive alpha, while the average beta component rises to 5.19%.
- **Economic interpretation.** Hedge funds still deliver returns, but much of what standard models call alpha is compensation for alternative beta exposures such as time-series momentum, variance selling, and carry.
- **Investor implication.** Sophisticated investors who can replicate these alternative factors value hedge funds much less than CAPM investors, and actual flows look closer to sophisticated-investor behavior.

2 Data and measurement

2.1 Return decomposition

The paper decomposes the average excess return of fund i into an alpha component and a beta component:

$$\mu_i = ac_i^* + bc_i^*. \tag{1}$$

The alpha component is the part based on private information, while the beta component is the part generated by publicly replicable factor strategies.

Interpretation: If the benchmark model omits hedge-fund strategies, alpha absorbs omitted beta premia. The empirical question is therefore not merely whether alpha is positive, but whether it remains positive after adding the right public strategy factors.

2.2 Hedge fund sample

- Monthly net-of-fee hedge fund returns are collected from January 1994 to December 2020.
- Four databases are combined to reduce selection and survivorship concerns: BarclayHedge, HFR, Morningstar, and TASS.
- The first 12 months of each fund's returns are removed to reduce backfill bias.

- The baseline selected sample contains 5,231 funds.
- The paper studies all funds and three broad investment styles: equity, macro, and arbitrage.
- Mutual funds are used as a comparison group for the time-trend exercise.

Interpretation: The sample is broad enough to study a distribution of performance, not just an average fund. That is crucial because hedge fund investors usually choose only a small number of funds, so dispersion matters.

2.3 Model set

The nine models are grouped as follows:

- **Reference model:** CAPM.
- **Standard models:** Carhart, Fama-French five-factor, Fung-Hsieh, and Asness-Moskowitz-Pedersen.
- **Machine-learning models:** two Kozak-Nagel-Santosh models trained on equity characteristic portfolios.
- **Additional-factor models:** JKKT and CP, which add economically motivated hedge-fund factors.

Interpretation: The point is not to find a perfect model. The paper instead compares imperfect models and asks which ones are less misspecified for hedge funds.

3 Models and empirical specifications

3.1 True model

A conceptual true model is

$$r_{i,t} = \alpha_i^* + b_i^{*\prime} f_t + \varepsilon_{i,t}, \tag{2}$$

with

$$ac_i^* = \alpha_i^*, \qquad bc_i^* = b_i^{*\prime} \lambda. \tag{3}$$

Interpretation: Here f_t includes all public strategies available to sophisticated investors. If these are fully included, alpha is true private-information value added.

3.2 Misspecified candidate model

For candidate model k,

$$r_{i,t} = \alpha_i^k + b_{i,I}^{k\prime} f_{I,t}^k + \varepsilon_{i,t}^k, \tag{4}$$

where $f_{I,t}^k$ are included factors and $f_{O,t}^k$ are omitted factors. The empirical components are

$$\widehat{ac}_i^k = \widehat{\alpha}_i^k, \qquad \widehat{bc}_i^k = \widehat{\mu}_i - \widehat{\alpha}_i^k, \qquad \widehat{bc}_{i,j}^k = \widehat{b}_{i,I,j}^k \widehat{\lambda}_{I,j}^k. \tag{5}$$

Interpretation: The beta component is defined as average return minus alpha. Therefore, when alpha falls after adding factors, the same return is being reclassified as beta.

3.3 Distributional objects

For each model, the paper estimates the cross-sectional distributions

$$\phi_{ac}^k, \qquad \phi_{bc}^k, \qquad \phi_{bc,j}^k. \tag{6}$$

The main characteristics are the mean M_1^k, standard deviation M_2^k, proportions $P^k(a)$ below a cutoff, and quantiles $Q^k(u)$.

Interpretation: This is more informative than average alpha. It identifies how many funds are positive-alpha funds, how dispersed performance is, and how much fund selection risk remains.

3.4 Model comparison

The paper compares the alpha distribution under model k with the CAPM reference model. A model that truly captures alternative strategies should reduce alpha relative to CAPM.

$$\Delta C = C^k - C^{CAPM}, \tag{7}$$

where C is a distribution characteristic such as mean alpha or the share of positive-alpha funds.

Interpretation: Under misspecification, cross-sectional information does not remove all estimation noise because omitted factors are common shocks across funds. The tests therefore use asymptotic theory appropriate for a large cross-section with common omitted-factor noise.

3.5 Sophisticated versus unsophisticated investor valuation

For an investor with stochastic discount factor m_t^k spanned by model k,

$$\alpha_i^k = (1 + r_f)E[m_t^k r_{i,t}]. \tag{8}$$

Interpretation: CP alpha is the value perceived by an investor who can replicate CP factors. CAPM alpha is the value perceived by a less sophisticated investor who only prices market exposure. The gap measures how much "exotic beta" is mistaken for alpha.

4 Identification and empirical design

- The empirical design is a model-comparison exercise, not a randomized treatment design.
- Identification relies on contrasting how different factor models reallocate the same fund returns between alpha and beta.
- The CAPM is used as the reference because it controls only for market exposure and is deliberately weak at capturing hedge-fund strategies.
- The formal tests adjust for the fact that omitted factors induce common estimation error across a large cross-section of funds.
- Robustness exercises address factor trading costs, alternative bias filters, investment-style categories, and time variation.

Interpretation: The key credibility feature is that the paper does not assume a model is correct. It compares misspecified models and asks which model is demonstrably less bad.

Table 2
Summary Statistics for the Hedge Fund Factors.
This table provides summary statistics for the factors used in the construction of the nine hedge fund models. Panel A reports the statistics for the US equity market and the standard factors used for performance evaluation. The size, value, momentum, investment, and profitability factors are computed using US equity data. The global value and momentum factors are constructed across multiple international asset classes. The term and default factors are computed using US bond data. The bond, commodity, and currency straddles are computed using option data on bonds, commodities, and currencies. Panel B reports the statistics for the additional hedge fund factors. The illiquidity and betting-against-beta (BAB) factors are computed using US equity data. The variance factor is computed using index option data on the S&P500. The carry and time-series (TS) momentum factors are constructed across multiple international asset classes. We define the straddle and variance factors as short positions to obtain positive premia. We report the annualized mean and standard deviation, skewness, kurtosis, and quantiles at 10% and 90% of the excess returns of the factors. The statistics are computed using monthly data between January 1994 and December 2020.

Panel A: Market and Standard Factors

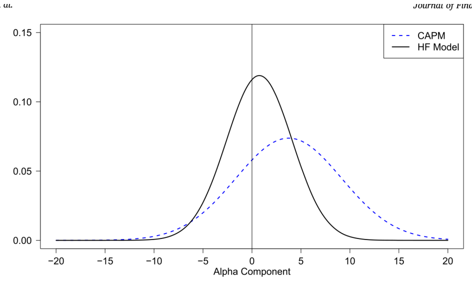
	Moments				Quantiles	
	Mean	Std Dev.	Skewness	Kurtosis	10%	90%
Market	8.81	15.48	-0.64	4.26	-5.15	6.02
Size	1.61	10.83	0.38	7.54	-3.49	3.69
Value	0.20	10.88	0.05	5.84	-3.30	3.47
Momentum	4.67	17.15	-1.42	12.88	-5.07	5.50
Investment	2.11	7.04	0.67	5.09	-2.14	2.70
Profitability	3.77	9.38	-0.47	13.49	-2.03	2.90
Global Value	1.30	6.19	-0.64	12.25	-1.72	1.76
Global Momentum	3.26	7.55	-0.30	5.46	-2.25	2.69
Term	-0.18	0.89	-0.03	4.22	-0.35	0.29
Default	0.01	0.77	1.90	17.66	-0.20	0.19
Straddle on Bonds	16.00	57.86	-1.85	8.99	-19.68	17.73
Straddle on Commodities	2.99	50.38	-1.32	6.07	-19.71	15.74
Straddle on Currencies	11.27	69.02	-1.56	6.64	-23.36	20.15

Panel B: Additional Factors

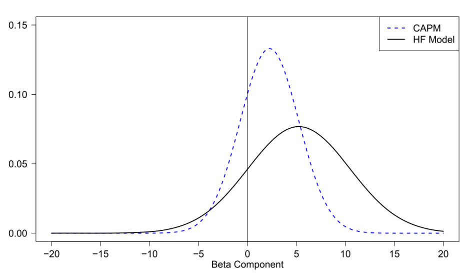
	Moments				Quantiles	
	Mean	Std Dev.	Skewness	Kurtosis	10%	90%
Illiquidity	6.52	12.97	-0.27	4.49	-3.60	4.99
BAB	8.58	13.77	-0.36	6.10	-3.58	5.21
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

5 Tables and figures

5.1 Figure 1 and Tables 1–2: the intuition and the data environment



(a) Distribution of the Alpha Components



(b) Distribution of the Beta Components

ibutions of the Alpha and Beta Components – A Simple Example.
ompares the average return decomposition obtained with a candidate hedge fund (HF) model and the CAPM. Panel A plots the cros
alpha components (annualized) under both models. In this simple example, the average fund returns are explained by four factors (tl

Table 1
Summary Statistics for the Equal-Weighted Portfolio of Hedge Funds.
This table provides summary statistics for the equal-weighted portfolio of all existing funds at the start of each month in the entire population and three investment categories: (i) equity funds (long-short, market neutral), (ii) macro funds (global macro, managed futures), and (iii) arbitrage funds (relative value, event driven). We report the annualized mean and standard deviation, skewness, kurtosis, and quantiles at 10% and 90% of the portfolio excess return. The statistics are computed using monthly data between January 1994 and December 2020.

	Moments				Quantiles	
	Mean	Std Dev.	Skewness	Kurtosis	10%	90%
All Funds	5.45	5.49	-0.31	4.51	-1.58	2.34
Equity	6.73	8.70	-0.47	5.32	-2.56	3.30
Long-Short	7.24	9.68	-0.44	5.31	-2.90	3.68
Market Neutral	3.07	2.72	-0.58	5.75	-0.66	1.13
Macro	4.40	6.12	0.40	3.57	-1.68	2.59
Global Macro	5.05	5.79	0.24	3.30	-1.66	2.50
Managed Futures	4.09	6.73	0.47	3.74	-1.79	2.92
Arbitrage	5.40	4.96	-2.16	14.57	-0.97	1.83
Relative Value	4.90	4.47	-2.46	18.03	-0.78	1.60
Event Driven	6.08	6.11	-1.74	11.79	-1.30	2.35

ıose reported by Getmansky et al. (2015) between

form models with the highest out-of-sample ability
erage returns of 50 characteristic-based equity portfo

(a) Figure 1: simple misspecification example

(b) Table 1: hedge fund return summary statistics

Read:

- Figure 1 illustrates that a model with fewer omitted factors produces lower measured alpha and higher measured beta.

- Table 1 shows that the equal-weighted hedge fund portfolio earns 5.45% per year with 5.49% volatility.
- Equity funds have the highest average return, 6.73%, but also high volatility, 8.70%.
- Arbitrage funds have strong negative skewness and high kurtosis, indicating crash-like left-tail risk.
- Table 2 shows that the additional factors have large premia: variance has a 38.10% annual mean, TS momentum 11.11%, BAB 8.58%, carry 6.85%, and illiquidity 6.52%.

Interpretation: Figure 1 is the conceptual heart of the paper: the same average return can look like alpha under CAPM but beta under a richer hedge-fund model. Tables 1–2 show why this is empirically important: hedge funds have meaningful average returns and alternative factors have large positive premia that can mechanically inflate alpha if omitted.

Economic message: The first lesson is that hedge funds operate in a world where alternative risk premia are economically large. A performance model that ignores these premia will overstate skill.

5.2 Tables 3–6: model set, misspecification, formal tests, and decomposition

Table 3
The Set of Hedge Fund Models.
This table summarizes the set of nine hedge fund models chosen for the empirical analysis. This set includes the CAPM as reference model as well as three distinct groups. The first group includes the standard models used in previous work, namely the Carhart, Five-Factor, Fung-Hsieh, and Asness-Moskowitz-Pedersen (AMP) models. The second group includes the two machine learning models of Kozak, Nagel, and Shantosh (KNS) formed with five characteristic-based equity portfolios and five principal components of these portfolios. The final group includes two models formed with the additional hedge fund factors. The first one is the model of Joenväärä et al. (2021) (JKKT). The second one combines the factors proposed by Carhart et al. (2014) and Pedersen (2018) (CP).

Model	List of Included Factors
Reference Model	
CAPM	Market
Standard Models	
Carhart	Market, Size, Value, Momentum
Five-Factor	Market, Size, Value, Investment, Profitability
Fung-Hsieh	Market, Size, Term, Default, Straddles (Bonds, Commodities, Currencies)
AMP	Market, Global Value and Momentum
Machine Learning Models	

(a) Table 3: nine factor models

JKKT	Market, Size, Value, Momentum, Illiquidity, BAB, TS Momentum
CP	Market, Size, Illiquidity, BAB, Variance, Carry, TS Momentum

Table 4
Model Misspecification Statistics.
This table provides misspecification statistics for the CAPM, the four standard models (Carhart, Five-Factor, Fung-Hsieh, AMP), the two machine learning models (KNS1, KNS2), and the two models with the additional factors (JKKT and CP). For each model, we measure the relative importance of the three components of the average fund return: (i) the alpha component (Alpha), (ii) the beta component due to the market (Beta Mkt), and the beta component due to the non-market factors (Beta Non-Mkt). These proportions, which are averaged across all funds, sum up to 100%. We also compute the average adjusted R^2 of the time-series regression of the fund return on the factor returns. We conduct this analysis for the entire hedge fund population (first four columns) and for the entire mutual fund population (last four columns).

	Hedge Funds				Mutual Funds			
	Relative Importance (%)				Relative Importance (%)			
	Alpha	Beta Mkt	Beta Non-Mkt	R^2	Alpha	Beta Mkt	Beta Non-Mkt	R^2
CAPM	52.75	47.25	0.00	20.43	-15.23	115.23	0.00	81.34
Carhart	47.14	46.60	6.26	25.28	-17.16	110.57	6.58	88.98
Five-Factor	49.04	46.65	4.31	24.85	-16.38	110.27	6.11	88.99
Fung-Hsieh	54.10	40.56	5.34	30.24	-14.96	107.78	7.18	85.88
AMP	47.08	47.37	5.55	24.36	-15.37	115.01	0.36	84.55
KNS1	52.52	48.24	-0.76	24.47	-17.41	114.58	2.84	83.40
KNS2	64.01	51.03	-15.04	26.39	-6.27	115.99	-9.73	87.09
JKKT	17.97	44.28	37.75	31.01	-20.01	110.20	9.81	89.65
CP	6.64	41.19	52.17	30.26	-24.33	109.06	15.27	87.15

and positive-alpha funds, and quantiles at 10% and 90%. To compute the standard deviation of the estimated differences, we replace T with

procedure, we conclude that the standard models are no better th… CAPM at capturing alternative hedge fund strategies.

(b) Table 4: misspecification statistics

Table 5
Formal Model Comparisons.
This table compares the CAPM with the four standard models (Carhart, Five-Factor, Fung-Hsieh, AMP), the two machine learning models (KNS1, KNS2), and the two models with the additional factors (JKKT and CP). We compute the differences in characteristics between the cross-sectional distributions of the alpha components under the CAPM and each model. We report the differences in the annualized mean and standard deviation, the proportions of funds with negative and positive alphas, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated differences. ***, **, * indicate that the null hypothesis of equal characteristics is rejected at the 1%, 5%, and 10% levels.

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Carhart	-0.31 (0.40)	-0.19 (0.31)	2.27 (2.50)	-2.27 (2.50)	-0.02 (0.40)	-0.60* (0.34)
Five-Factor	-0.21 (0.46)	0.02 (0.34)	2.41 (3.01)	-2.41 (3.01)	-0.01 (0.50)	-0.18 (0.40)
Fung-Hsieh	0.08 (0.60)	-0.20 (0.35)	0.52 (3.88)	-0.52 (3.88)	0.28 (0.56)	-0.15 (0.47)
AMP	-0.31 (0.42)	0.06 (0.37)	2.50 (2.44)	-2.50 (2.44)	-0.17 (0.37)	-0.34 (0.45)
KNS1	-0.01 (0.39)	0.43 (0.29)	1.59 (1.78)	-1.59 (1.78)	-0.26 (0.29)	0.19 (0.38)
KNS2	0.63 (0.49)	0.20 (0.31)	-2.47 (2.65)	2.47 (2.65)	0.58 (0.41)	0.65 (0.47)
JKKT	-1.93*** (0.67)	0.41 (0.50)	14.76*** (4.01)	-14.76*** (4.01)	-2.32*** (0.66)	-1.98*** (0.68)
CP	-2.56*** (0.76)	2.13*** (0.50)	19.82*** (4.27)	-19.82*** (4.27)	-4.22*** (0.73)	-1.42* (0.76)

Table 6
Decomposition of Average Fund Returns.
This table shows the decomposition of average fund returns under the CAPM, the four standard models (Carhart, Five-Factor, Fung-Hsieh, AMP), the two machine learning models (KNS1, KNS2), and the two models

(a) Table 5: formal model comparisons

the proportions of funds with negative and positive alphas, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panel B reports the characteristics of the cross-sectional distribution of beta components under each model.

Panel A: Distribution of the Alpha Components

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
CAPM	2.93 (0.94)	7.01 (0.48)	27.15 (5.49)	72.85 (5.49)	-3.95 (0.67)	10.06 (0.67)
Carhart	2.62 (0.84)	6.82 (0.31)	29.42 (5.08)	70.58 (5.08)	-3.97 (0.57)	9.46 (0.55)
Five-Factor	2.72 (0.89)	7.03 (0.31)	29.55 (5.23)	70.45 (5.23)	-3.96 (0.63)	9.87 (0.54)
Fung-Hsieh	3.01 (0.74)	6.81 (0.29)	27.66 (3.60)	72.34 (3.60)	-3.67 (0.47)	9.90 (0.51)
AMP	2.62 (0.92)	7.08 (0.28)	29.65 (5.62)	70.35 (5.62)	-4.12 (0.59)	9.71 (0.54)
KNS1	2.92 (0.87)	7.44 (0.42)	28.73 (5.13)	71.27 (5.13)	-4.21 (0.63)	10.24 (0.60)
KNS2	3.56 (0.86)	7.21 (0.37)	24.68 (4.33)	75.32 (4.33)	-3.37 (0.55)	10.70 (0.62)
JKKT	1.00 (0.73)	7.42 (0.31)	41.90 (4.80)	58.10 (4.80)	-6.27 (0.60)	8.07 (0.38)
CP	0.37 (0.86)	9.15 (0.40)	46.97 (5.31)	53.03 (5.31)	-8.17 (0.77)	8.64 (0.44)

Panel B: Distribution of the Beta Components

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
CAPM	2.62 (0.83)	4.37 (0.67)	22.54 (9.57)	77.46 (9.57)	-0.71 (0.61)	8.01 (0.73)
Carhart	2.94 (0.81)	4.30 (0.52)	17.99 (5.68)	82.01 (5.68)	-0.65 (0.42)	8.20 (0.72)
Five-Factor	2.83 (0.84)	4.48 (0.51)	20.61 (5.62)	79.39 (5.62)	-1.06 (0.41)	8.32 (0.75)
Fung-Hsieh	2.55 (0.78)	4.39 (0.52)	22.58 (5.98)	77.42 (5.98)	-1.23 (0.44)	7.96 (0.67)
AMP	2.94 (0.84)	4.66 (0.49)	18.98 (5.41)	81.02 (5.41)	-0.96 (0.43)	8.56 (0.73)
KNS1	2.64 (0.80)	5.14 (0.57)	23.04 (5.71)	76.96 (5.71)	-1.37 (0.56)	8.38 (0.86)
KNS2	2.00 (0.82)	4.47 (0.49)	27.53 (7.96)	72.47 (7.96)	-1.61 (0.59)	7.28 (0.69)
JKKT	4.56 (0.83)	5.90 (0.42)	13.90 (2.20)	86.10 (2.20)	-0.50 (0.28)	11.12 (0.74)
CP	5.19 (0.93)	7.64 (0.49)	15.91 (2.17)	84.09 (2.17)	-1.17 (0.35)	12.78 (1.11)

the JKKT and CP models are better equipped to capture alternative strategies and sharpen the fund return decomposition.

Both the CAPM and the standard models produce large alpha … ponents. Panel A shows that the average alpha clusters around

(b) Table 6: alpha and beta distributions

Read:

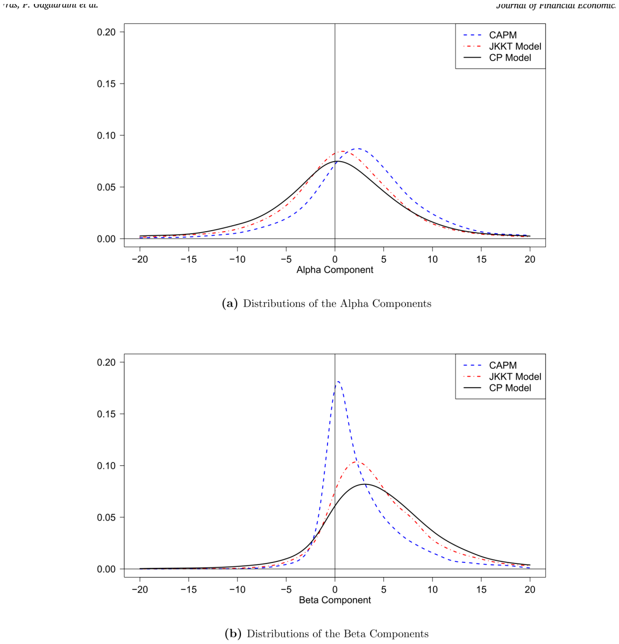
- Table 3 organizes the models into reference, standard, machine-learning, and additional-factor models.

- Table 4 shows low hedge-fund adjusted R^2 values, between 20.43% and 31.01%, leaving room for omitted factors.
- In Table 4, CP attributes only 6.64% of the average hedge-fund return to alpha and 52.17% to non-market beta.
- Table 5 shows that standard and machine-learning models are not significantly different from the CAPM, but JKKT and CP are.
- In Table 6, CAPM average alpha is 2.93% and 72.85% of funds have positive alpha.
- Under CP, average alpha falls to 0.37%, while the average beta component rises to 5.19% and is positive for 84.09% of funds.

Interpretation: These tables overturn the usual alpha story. The standard models create a picture in which hedge funds deliver high alpha while remaining mostly immune to non-market systematic risk. The CP model reverses that interpretation: the average hedge fund return is mostly compensation for alternative beta exposures.

Economic message: The economic message is not that hedge funds create no value. Rather, the value attribution changes. What investors call "skill" depends on which strategies they can replicate outside the fund.

5.3 Figure 2 and Table 7: the CP model and economically important factors



(a) Distributions of the Alpha Components

(b) Distributions of the Beta Components

butions of the Alpha and Beta Components.

(a) Figure 2: alpha/beta distributions under CAPM, JKKT, and CP

Table 7
Economic Importance of the Hedge Fund Factors.
This table measures the economic importance of each factor in the CP model as a driver of average fund returns. Panel A reports the characteristics of the cross-sectional distribution of the beta components due to each factor. We report the annualized mean and standard deviation, the proportions of funds with negative and positive contributions, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panel B reports the cross-sectional correlation between the beta components for each pair of factors.

Panel A: Distribution of the Beta Components for Each Factor

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Market	2.29 (1.27)	3.84 (1.19)	21.62 (6.13)	78.38 (6.13)	-0.55 (0.15)	6.99 (2.05)
Size	0.23 (0.26)	1.01 (0.32)	38.23 (4.26)	61.77 (4.26)	-0.31 (0.03)	1.02 (0.53)
Illiquidity	0.05 (0.07)	1.26 (0.29)	44.94 (3.25)	55.06 (3.25)	-0.75 (0.24)	0.85 (0.38)
Betting Against Beta	0.35 (0.25)	2.22 (0.69)	35.14 (3.80)	64.86 (3.80)	-1.30 (0.44)	2.23 (0.90)
Variance	0.78 (0.25)	5.17 (0.70)	36.74 (1.89)	63.26 (1.89)	-3.08 (0.54)	4.53 (0.87)
Carry	0.43 (0.17)	2.56 (0.46)	39.21 (2.38)	60.79 (2.38)	-1.89 (0.37)	2.73 (0.51)
Time-Series Momentum	1.06 (0.32)	4.03 (0.78)	42.86 (1.51)	57.14 (1.51)	-1.35 (0.34)	4.49 (0.72)

Panel B: Correlations Between the Beta Components

	Size	Illiquidity	BAB	Variance	Carry	TS Mom
Market	-0.08	-0.01	-0.01	-0.03	-0.02	-0.18
Size		-0.07	-0.00	-0.06	0.06	0.05
Illiquidity			-0.09	0.14	-0.02	0.08
Betting Against Beta				-0.07	0.03	-0.05
Variance					0.04	-0.09
Carry						-0.13

erage alphas and lower proportions of positive-alpha population and for all investment categories). For this ily focus on the CP model for the rest of our analysis.

arbitrage funds suffer from less effective hedging strategies funding constraints (Buraschi et al., 2014b).

At the same time, some factors in the CP model cater styles. For instance, TS momentum plays a key role for macro

(b) Table 7: factor-by-factor beta components

Read:

- Figure 2 shows that CP shifts the alpha distribution toward zero and the beta distribution away from zero.

- Table 7 decomposes CP beta by factor. The market contribution averages 2.29% and is positive for 78.38% of funds.
- The most important non-market factors are TS momentum at 1.06%, variance at 0.78%, and carry at 0.43%.
- The majority of funds have positive exposure contributions to each of the CP factors.
- Pairwise correlations among factor beta components are small, ranging roughly from -0.18 to 0.14.

Interpretation: The low correlations in Table 7 are important: CP is not merely repackaging one common hedge-fund exposure. Different funds use different combinations of market, size, variance-selling, carry, BAB, illiquidity, and trend-following strategies.

Economic message: Hedge fund beta is multidimensional. A single style benchmark cannot capture it well, and even category-level benchmarks can miss fund-level heterogeneity.

5.4 Tables 8–9: investment styles and style-specific factors

Table 8
Decomposition of Average Fund Returns – Investment Categories.
This table shows the decomposition of average fund returns under the CAPM and the two models with the additional factors (JKKT and CP) across investment styles. Panel A reports the characteristics of the cross-sectional distributions of the alpha and beta components across equity funds (long-short, market neutral). We report the annualized mean and standard deviation, the proportions of funds with negative and positive alphas, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panels B and C repeat the analysis for macro funds (global macro, managed futures) and arbitrage funds (relative value, event driven).

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Panel A: Equity Funds						
Distribution of the Alpha Components						
CAPM	2.40 (1.06)	7.09 (0.48)	30.23 (6.31)	69.77 (6.31)	-4.52 (0.83)	9.54 (0.96)
JKKT	1.12 (0.81)	6.98 (0.31)	42.98 (5.48)	57.02 (5.48)	-5.73 (0.75)	7.77 (0.66)
CP	0.58 (0.96)	9.08 (0.49)	47.38 (5.46)	52.62 (5.46)	-7.84 (1.02)	8.53 (0.79)
Distribution of the Beta Components						
CAPM	4.18 (1.16)	5.19 (0.70)	12.94 (8.25)	87.06 (8.25)	-0.14 (0.74)	10.66 (1.39)
JKKT	5.46 (1.18)	5.75 (0.54)	10.59 (2.37)	89.41 (2.37)	-0.10 (0.43)	12.19 (1.41)
CP	6.00 (1.31)	7.58 (0.61)	13.69 (2.77)	86.31 (2.77)	-0.75 (0.56)	13.71 (1.77)
Panel B: Macro Funds						
Distribution of the Alpha Components						
CAPM	3.71 (1.72)	8.04 (0.78)	25.73 (6.47)	74.27 (6.47)	-4.26 (0.89)	12.41 (2.14)
JKKT	-0.14 (1.42)	9.14 (0.65)	51.52 (7.66)	48.48 (7.66)	-9.25 (1.43)	8.70 (0.75)
CP	-0.39 (1.65)	11.04 (0.70)	51.83 (7.02)	48.17 (7.02)	-10.91 (1.58)	10.26 (1.00)
Distribution of the Beta Components						
CAPM	1.01 (1.08)	3.70 (0.68)	43.44 (21.78)	56.56 (21.78)	-1.76 (1.29)	5.44 (0.66)
JKKT	4.87 (1.11)	7.49 (0.74)	20.32 (3.33)	79.68 (3.33)	-1.57 (0.44)	12.86 (1.30)
CP	5.11 (1.33)	9.62 (0.79)	23.74 (3.89)	76.26 (3.89)	-2.60 (0.63)	14.18 (1.35)
Panel C: Arbitrage Funds						
Distribution of the Alpha Components						
CAPM	2.81 (1.24)	5.61 (0.44)	24.74 (9.71)	75.26 (9.71)	-2.79 (1.31)	8.67 (0.63)
JKKT	1.97 (0.89)	5.70 (0.29)	31.03 (8.00)	68.97 (8.00)	-3.66 (0.92)	7.60 (0.50)
CP	0.87 (1.06)	6.81 (0.44)	41.64 (9.30)	58.36 (9.30)	-5.48 (1.20)	7.62 (0.49)

(a) Table 8: alpha/beta decomposition across styles

Table 9
Economic Importance of the Hedge Fund Factors – Investment Categories.
This table measures the economic importance of each factor in the CP model as a driver of average fund returns across investment styles. Panel A reports the characteristics of the cross-sectional distributions of the beta components due to each factor across equity funds (long-short, market neutral). We report the annualized mean and standard deviation, the proportions of funds with negative and positive contributions, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panel B and C repeat the analysis for macro funds (global macro, managed futures) and arbitrage funds (relative value, event driven).

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Panel A: Equity Funds						
Market	3.67 (2.07)	4.53 (1.22)	14.54 (8.16)	85.46 (8.16)	-0.19 (0.61)	9.33 (3.18)
Size	0.42 (0.57)	1.32 (0.42)	32.68 (11.70)	67.32 (11.70)	-0.35 (0.11)	1.65 (1.04)
Illiquidity	0.14 (0.10)	1.49 (0.35)	44.13 (2.66)	55.87 (2.66)	-0.85 (0.25)	1.27 (0.48)
Betting Against Beta	0.25 (0.21)	2.66 (0.80)	38.48 (3.90)	61.52 (3.90)	-1.71 (0.55)	2.55 (1.05)
Variance	0.67 (0.25)	5.06 (0.70)	38.58 (2.19)	61.42 (2.19)	-3.12 (0.54)	4.46 (0.85)
Carry	0.34 (0.17)	2.80 (0.46)	44.78 (2.67)	55.22 (2.67)	-2.11 (0.41)	2.73 (0.52)
Time-Series Momentum	0.51 (0.17)	2.83 (0.55)	42.08 (2.04)	57.92 (2.04)	-1.56 (0.46)	3.05 (0.74)
Panel B: Macro Funds						
Market	1.18 (0.62)	3.47 (1.10)	35.11 (4.40)	64.89 (4.40)	-1.40 (0.35)	4.87 (1.62)
Size	0.07 (0.09)	0.83 (0.20)	50.59 (9.02)	49.41 (9.02)	-0.43 (0.28)	0.64 (0.20)
Illiquidity	-0.08 (0.10)	1.26 (0.31)	54.07 (3.37)	45.93 (3.37)	-0.90 (0.37)	0.75 (0.28)
Betting Against Beta	0.21 (0.19)	1.99 (0.65)	40.83 (4.63)	59.17 (4.63)	-1.30 (0.47)	1.84 (0.76)
Variance	0.22 (0.35)	6.56 (0.87)	49.41 (3.11)	50.59 (3.11)	-4.73 (0.92)	4.55 (0.76)
Carry	0.41 (0.26)	2.79 (0.57)	41.89 (3.33)	58.11 (3.33)	-2.43 (0.58)	3.23 (0.64)
Time-Series Momentum	3.10 (1.11)	5.83 (1.23)	24.43 (3.26)	75.57 (3.26)	-0.73 (0.12)	10.22 (2.20)
Panel C: Arbitrage Funds						
Market	1.68 (0.90)	2.52 (0.75)	16.96 (6.38)	83.04 (6.38)	-0.19 (0.21)	4.40 (1.44)
Size	0.16 (0.20)	0.60 (0.17)	32.82 (8.47)	67.18 (8.47)	-0.11 (0.04)	0.56 (0.45)
Illiquidity	0.07 (0.12)	0.87 (0.18)	36.89 (7.20)	63.11 (7.20)	-0.40 (0.06)	0.60 (0.32)
Betting Against Beta	0.61 (0.39)	1.76 (0.58)	25.35 (4.61)	74.65 (4.61)	-0.68 (0.14)	2.15 (0.95)
Variance	1.47 (0.42)	3.34 (0.55)	21.90 (2.39)	78.10 (2.39)	-0.78 (0.08)	4.59 (0.97)
Carry	0.56 (0.20)	1.91 (0.34)	29.67 (3.66)	70.33 (3.66)	-0.86 (0.17)	2.37 (0.52)
Time-Series Momentum	-0.30 (0.19)	1.53 (0.37)	62.12 (4.12)	37.88 (4.12)	-1.64 (0.57)	0.82 (0.17)

(b) Table 9: factor components across styles

Read:

- Table 8 shows that CP reduces average alpha in every major style.
- CP alpha is about 0.6% for equity funds, -0.4% for macro funds, and 0.9% for arbitrage funds, compared with CAPM alphas of 2.4%, 3.7%, and 2.8%.
- Cross-sectional dispersion remains large: CP alpha volatility ranges from 6.8% to 11.0% across styles, and beta volatility ranges from 4.9% to 9.6%.
- Table 9 shows that carry is positive in all three styles, while TS momentum is especially important for macro funds.
- Variance also matters across styles, especially for equity and arbitrage funds.

Interpretation: Style labels are informative but incomplete. Two macro funds can have very different alpha and beta components, and two arbitrage funds can have different exposures to BAB, carry, and variance. This limits the usefulness of peer benchmarking that gives every fund in a style the same benchmark.

Economic message: The CP factors partly transcend style boundaries. Carry, trend-following, and variance-selling are not isolated to one category; they are broad hedge-fund technologies.

5.5 Table 10 and Figures 3–4: fund characteristics and time variation

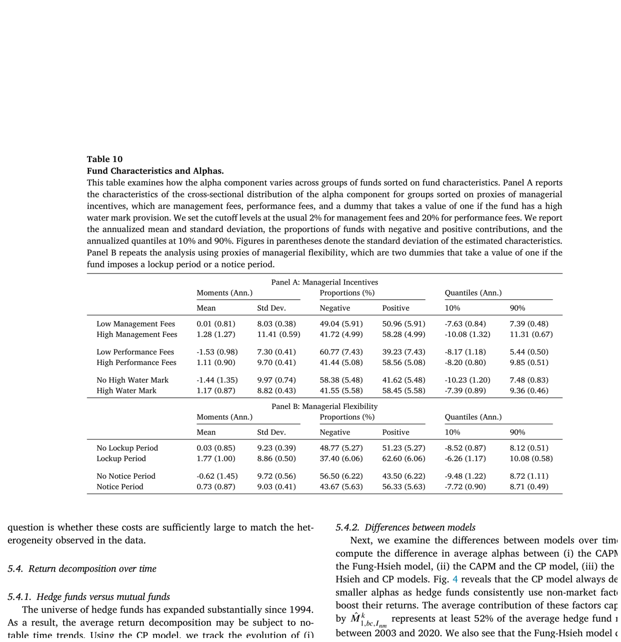


Table 10
Fund Characteristics and Alphas.
This table examines how the alpha component varies across groups of funds sorted on fund characteristics. Panel A reports the characteristics of the cross-sectional distribution of the alpha component for groups sorted on proxies of managerial incentives, which are management fees, performance fees, and a dummy that takes a value of one if the fund has a high water mark provision. We set the cutoff levels at the usual 2% for management fees and 20% for performance fees. We report the annualized mean and standard deviation, the proportions of funds with negative and positive contributions, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panel B repeats the analysis using proxies of managerial flexibility, which are two dummies that take a value of one if the fund imposes a lockup period or a notice period.

Panel A: Managerial Incentives

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Low Management Fees	0.01 (0.81)	8.03 (0.38)	49.04 (5.91)	50.96 (5.91)	-7.63 (0.84)	7.39 (0.48)
High Management Fees	1.28 (1.27)	11.41 (0.59)	41.72 (4.99)	58.28 (4.99)	-10.08 (1.32)	11.31 (0.67)
Low Performance Fees	-1.53 (0.98)	7.30 (0.41)	60.77 (7.43)	39.23 (7.43)	-8.17 (1.18)	5.44 (0.50)
High Performance Fees	1.11 (0.90)	9.70 (0.41)	41.44 (5.08)	58.56 (5.08)	-8.20 (0.80)	9.85 (0.51)
No High Water Mark	-1.44 (1.35)	9.97 (0.74)	58.38 (5.48)	41.62 (5.48)	-10.23 (1.20)	7.48 (0.83)
High Water Mark	1.17 (0.87)	8.82 (0.43)	41.55 (5.58)	58.45 (5.58)	-7.39 (0.89)	9.36 (0.46)

Panel B: Managerial Flexibility

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
No Lockup Period	0.03 (0.85)	9.23 (0.39)	48.77 (5.27)	51.23 (5.27)	-8.52 (0.87)	8.12 (0.51)
Lockup Period	1.77 (1.00)	8.86 (0.50)	37.40 (6.06)	62.60 (6.06)	-6.26 (1.17)	10.08 (0.58)
No Notice Period	-0.62 (1.45)	9.72 (0.56)	56.50 (6.22)	43.50 (6.22)	-9.48 (1.22)	8.72 (1.11)
Notice Period	0.73 (0.87)	9.03 (0.41)	43.67 (5.63)	56.33 (5.63)	-7.72 (0.90)	8.71 (0.49)

question is whether these costs are sufficiently large to match the heterogeneity observed in the data.

5.4. Return decomposition over time

5.4.1. Hedge funds versus mutual funds

The universe of hedge funds has expanded substantially since 1994. As a result, the average return decomposition may be subject to notable time trends. Using the CP model, we track the evolution of (i)

5.4.2. Differences between models

Next, we examine the differences between models over tim compute the difference in average alphas between (i) the CAPM the Fung-Hsieh model, (ii) the CAPM and the CP model, (iii) the Hsieh and CP models. Fig. 4 reveals that the CP model always de smaller alphas as hedge funds consistently use non-market fact boost their returns. The average contribution of these factors cap by $\hat{M}^{k}_{1,bc,I_{nm}}$ represents at least 52% of the average hedge fund r between 2003 and 2020. We also see that the Fung-Hsieh model

(a) Table 10: alpha by incentives and flexibility

(b) Figure 3: time variation in return components

Read:

- Table 10 shows that high-incentive or more flexible funds are more likely to have positive alpha.
- High performance-fee funds have average alpha of 1.11% and 58.56% positive-alpha funds; low performance-fee funds have -1.53% and 39.23%.
- Funds with lockup periods have average alpha of 1.77% and 62.60% positive-alpha funds, compared with 0.03% and 51.23% for funds without lockups.
- Figure 3 shows convergence between hedge funds and mutual funds: the hedge-fund alpha gap narrows to 1.6% by 2020.
- Hedge funds also load more on the equity market after the financial crisis; by 2020 the market beta contribution reaches 2.3% per year.
- Figure 4 shows that CP always produces lower average alpha than CAPM and Fung-Hsieh.

Interpretation: Table 10 suggests that fund characteristics can help investors screen for positive alpha, but the differences are not clean enough to remove selection uncertainty. Figure 3 shows that the hedge fund industry becomes more mutual-fund-like over time, while Figure 4 shows that richer factor models consistently reclassify alpha as beta.

Economic message: Incentives and flexibility matter, but they do not eliminate fund-selection risk. At the industry level, performance erodes and market exposure rises, so investors seeking hedge-fund alpha need both manager selection and beta hedging.

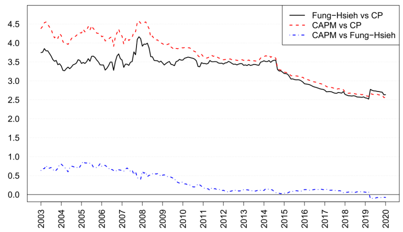


Table 11
Fund Flows and Investor Sophistication.
This table measures how investor flows contemporaneously respond to the different components of the average fund return under the CP model: (i) the alpha component, (ii) the beta component due to the market, and (iii) the beta component due to the non-market factors (size, illiquidity, BAB, variance, carry, TS momentum). For each of the non-overlapping five-year periods between 1996 and 2020, we measure the three components for all funds. We then rank them according to their average monthly net flows and group them into quintiles (low, 2, 3, 4, high). Panel A reports the characteristics of the cross-sectional distribution of the alpha components (pooled over all five-year periods) for the five flow groups. We report the annualized mean and standard deviation, the proportions of funds with negative and positive alphas, and the annualized quantiles at 10% and 90%. Figures in parentheses denote the standard deviation of the estimated characteristics. Panels B and C repeat the analysis for the beta components due to the market and the non-market factors.

Panel A: Distribution of the Alpha Components

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Low	-0.57 (1.54)	10.71 (0.79)	54.84 (5.88)	45.16 (5.88)	-10.53 (1.33)	9.55 (0.74)
2	0.39 (1.33)	9.03 (0.72)	47.82 (5.66)	52.18 (5.66)	-9.01 (0.98)	9.15 (0.83)
3	1.52 (1.32)	8.42 (0.76)	41.49 (6.10)	58.51 (6.10)	-7.18 (0.90)	10.33 (0.65)
4	2.92 (1.44)	8.85 (0.82)	34.57 (6.90)	65.43 (6.90)	-5.97 (1.18)	13.00 (0.56)
High	3.25 (1.24)	9.87 (0.61)	29.61 (5.31)	70.39 (5.31)	-4.92 (0.98)	13.18 (0.74)

Panel B: Distribution of the Beta Components (Market Factor)

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Low	2.32 (1.98)	4.19 (1.75)	18.19 (19.41)	81.81 (19.41)	-0.53 (0.84)	8.37 (3.00)
2	3.15 (2.45)	5.09 (1.75)	14.25 (18.87)	85.75 (18.87)	-0.17 (1.19)	10.34 (4.27)
3	2.91 (2.26)	4.94 (1.84)	15.85 (18.60)	84.15 (18.60)	-0.17 (1.13)	9.48 (4.06)
4	2.40 (1.84)	4.42 (1.74)	19.96 (15.33)	80.04 (15.33)	-0.33 (0.71)	8.29 (3.30)
High	1.95 (1.45)	3.90 (1.61)	23.07 (13.06)	76.93 (13.06)	-0.46 (0.45)	7.31 (2.63)

Panel C: Distribution of the Beta Components (Non-Market Factors)

	Moments (Ann.)		Proportions (%)		Quantiles (Ann.)	
	Mean	Std Dev.	Negative	Positive	10%	90%
Low	2.38 (0.90)	8.18 (3.58)	32.13 (8.54)	67.87 (8.54)	-4.43 (1.93)	10.44 (3.66)
2	2.13 (0.83)	7.36 (3.03)	35.93 (12.43)	64.07 (12.43)	-4.26 (1.59)	9.60 (3.57)
3	2.14 (0.81)	6.94 (3.28)	32.07 (11.95)	67.93 (11.95)	-4.23 (2.02)	9.34 (3.59)
4	2.30 (0.78)	7.16 (3.19)	34.30 (8.29)	65.70 (8.29)	-3.88 (1.79)	9.63 (3.59)
High	2.37 (0.81)	7.82 (3.09)	33.70 (6.23)	66.30 (6.23)	-3.81 (1.70)	8.65 (3.46)

availability

ıe authors do not have permission to share data.
ode and fake pseudodata are available at https://doi.org/10.5281/
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owledgements

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Appendix A. Supplementary material

5.6 Table 11: flows and investor sophistication

Read:

- Table 11 sorts funds by net flows and asks whether flows chase CP alpha, market beta, or non-market beta.
- High-flow funds have average alpha of 3.25% and 70.39% positive-alpha funds.
- Low-flow funds have average alpha of -0.57% and only 45.16% positive-alpha funds.
- Market and non-market beta components are not systematically larger for high-flow funds.
- The result suggests that real-world hedge fund investors respond more to CP alpha than to raw beta exposure.

Interpretation: The flow evidence is a direct test of investor sophistication. If investors were naive CAPM investors, they would chase funds with high alternative beta, because those returns look like alpha under CAPM. Instead, flows track CP alpha more closely, suggesting investors understand at least some exotic beta exposures.

Economic message: The market for hedge funds is not perfectly naive. Capital moves toward funds with higher model-adjusted alpha, although the overlap between low-flow and high-flow distributions shows that learning remains noisy.

6 Short summary

- The paper develops a method to decompose hedge fund returns into alpha and beta under factor-model misspecification.
- Misspecification matters because omitted alternative factors cause alpha to absorb beta premia.
- The method estimates entire distributions of alpha and beta, not only averages.
- It also provides formal model-comparison tests that account for common omitted-factor estimation noise.

- Standard hedge-fund models and equity-trained machine-learning models are not significantly better than CAPM for separating alpha and beta.
- CP, which includes market, size, illiquidity, BAB, variance, carry, and TS momentum, performs best.
- Under CP, average hedge fund alpha falls close to zero and beta becomes the main source of return.
- TS momentum, variance, and carry are the most important additional factors.
- Fund heterogeneity is large even within broad styles.
- High-incentive and more flexible funds are more likely to deliver positive alpha.
- Hedge fund performance increasingly resembles mutual fund performance over time.
- Actual investor flows look more consistent with sophisticated CP-based valuation than naive CAPM valuation.

7 Conclusion

7.1 Core conclusion

The central conclusion is that hedge fund performance cannot be evaluated by asking whether CAPM or standard-model alpha is positive. Because hedge funds use alternative strategies, standard alphas often mix true skill with systematic beta premia. Once CP-style factors are included, the average alpha falls from roughly 3% per year to close to zero, and only about half of funds remain positive-alpha funds.

7.2 Interpretation for alpha versus beta

The paper does not claim that hedge funds are useless. It shows that much of their return is replicable by public strategies. This distinction matters because investors should pay high fees only for the nonreplicable component. If a return can be produced with trend-following, carry, variance-selling, BAB, illiquidity, or size factors, then it is beta from the perspective of a sophisticated investor.

7.3 Implications for hedge fund investors

Investors should evaluate hedge funds using benchmarks that include realistic alternative factors. A manager who looks skilled under CAPM may simply be loading on carry or short variance. Conversely, the paper shows that some funds still deliver positive alpha after CP adjustment, so the investor problem becomes harder but not pointless: the goal is to identify managers with positive residual alpha after controlling for alternative beta.

7.4 Implications for fund selection and due diligence

The large cross-sectional dispersion in both alpha and beta means that investors cannot rely only on style labels. Due diligence should examine the actual return decomposition of each fund. Incentive and flexibility variables, such as performance fees, high-water marks, lockups, and notice periods, help screen funds, but they do not fully identify winners. The overlap between distributions remains large, which means fund selection remains noisy and costly.

7.5 Implications for benchmark design

Peer benchmarks are particularly problematic. If all macro funds or all arbitrage funds are assigned the same style benchmark, the benchmark hides large differences in factor exposures. A better benchmark system should be fund-specific or at least more granular, combining style information with measurable factor exposures to carry, time-series momentum, variance, BAB, illiquidity, and other alternative strategies.

7.6 Implications for active-management theory

The findings speak directly to models of active management. The Berk-Green logic predicts that competition and flows should push net alpha toward zero. The paper finds that average CP alpha is close to zero, but also that alpha heterogeneity remains large. This supports models in which search frictions, investor sophistication, and fund heterogeneity allow some dispersion in performance to persist.

7.7 Implications for regulation and systemic risk

For policymakers, the decomposition matters because hedge funds may look like low-beta alpha producers under conventional models while actually carrying systematic exposures to crowded strategies. If many funds share exposures to variance selling, carry, or momentum, stress can appear through factor unwinds rather than through conventional equity beta. Monitoring hedge fund risk therefore requires a broader factor map than the CAPM or equity-based models provide.

7.8 Limitations and future research

The CP model is less misspecified, not perfectly specified. Hedge funds may use dynamic strategies, nonlinear option positions, leverage, and private trades that are not captured fully by monthly linear factors. Future research could apply the method to narrower strategy categories, international funds, private funds, high-frequency fund exposures, or nonlinear factor models. Another open question is whether alternative factor premia continue to decline as more investors learn to replicate them.

7.9 Final takeaway

The paper's practical lesson is simple: hedge fund alpha is partly in the eye of the benchmark holder. For unsophisticated investors, exotic beta can look like alpha. For sophisticated investors, much of that alpha becomes beta. The best empirical evaluation therefore asks not only whether a fund earns high returns, but whether those returns survive a benchmark that reflects the strategies sophisticated investors can actually implement.